Harsha Vardhan G

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PROFESSIONAL PROFILE

Research Software Engineer with 4+ years of experience designing and shipping backend platforms, data pipelines, developer tools, full-stack applications, and privacy/security infrastructure. Strong end-to-end ownership across ambiguous requirements, architecture, implementation, deployment, performance tuning, and production support. Experienced with Python, Django REST Framework, Flask, Node.js, React, TypeScript, PostgreSQL, Firebase, WebSockets, Docker, AWS, and GCP. Technical depth in distributed workflows, real-time collaboration, cryptographic systems, secure data handling, and automation. Known for turning complex systems problems into maintainable products, mentoring engineers, and building tools that improve team velocity.

TECHNICAL SKILLS

Backend & Platforms: Python, Django REST Framework, Flask, Node.js, REST APIs, Microservices, PostgreSQL, Data Pipelines, API Design
Frontend & Product: React, TypeScript, JavaScript, Vite, CodeMirror, PDF.js, Responsive Web Applications, Developer UX
Distributed & Real-time Systems: WebSockets, Yjs, Real-time Collaboration, Event-driven Workflows, Relay Services, Monitoring
Infrastructure & Cloud: Docker, Linux, AWS, GCP, Firebase Auth/Firestore/Storage, CI/CD, Deployment Automation, Cloud Run
Data & Performance: PostgreSQL Optimization, Real-time Analytics, Data Processing, Query Tuning, Latency Reduction, Automation
Security & Privacy: Secure Data Handling, Access Control, End-to-End Encryption, AES-GCM, RSA-OAEP, PKI, ZKPs, Key Management
Leadership: Technical Design, End-to-End Ownership, Mentoring, Cross-functional Collaboration, Agile/Scrum, Production Support
Languages: Python, JavaScript/TypeScript, C/C++, Rust familiar, Go familiar, Solidity familiar

WORK EXPERIENCE

ADAPT Centre | Research Software Engineer | Dublin, Ireland *Jan 2025 - Present*

- Design and build secure backend systems, full-stack applications, and developer-facing tools for privacy-preserving digital infrastructure.
- Translate complex research, legal, privacy, and stakeholder requirements into architecture decisions, implementation plans, and production-grade code.
- Architect encryption, anonymization, and cryptographic verification workflows while balancing security guarantees, usability, and latency.
- Build and deploy production-facing utilities across React, TypeScript, Python, Firebase, WebSockets, Docker, and cloud hosting workflows.
- Collaborate with researchers, engineers, policy stakeholders, and open-source maintainers to deliver trustworthy public-facing technical systems.
- Own systems from prototype to deployment, including debugging, performance optimization, documentation, and operational hardening.

Stratforge | Software Engineer | Chennai, India *Sept 2020 - Aug 2023*

- Built and maintained backend data pipelines and REST APIs using Django REST Framework, PostgreSQL, and containerized microservices for enterprise SaaS systems.
- Developed automation tooling across deployment, monitoring, and data-processing workflows, reducing manual engineering effort and improving operational reliability.
- Co-architected RedFlare, a real-time analytics and investigative monitoring platform that automated data ingestion, processing, and issue analysis, improving issue resolution time by 20%.
- Optimized PostgreSQL queries and backend service paths to improve API response latency and support higher-throughput data workflows.
- Owned engineering problems end to end, from requirements gathering and design through implementation, testing, deployment, observability, and production support.
- Mentored 4+ junior engineers on secure coding practices, backend architecture, debugging, and production-readiness standards.

SELECTED PROJECTS

ZKP-Based Privacy Framework on Zcash | *ADAPT Centre*

- Designed and implemented an end-to-end privacy-first voting framework using Zcash shielded addresses, zk-SNARKs, and strict application-layer data handling controls.
- Built backend workflows that preserved voter anonymity while enabling cryptographic verification, auditability, and secure transaction handling.
- Converted privacy and compliance requirements into concrete system behavior, reducing unnecessary exposure of sensitive user data.

InvizCrypt | *ADAPT Centre*

- Built and deployed a browser-based collaborative LaTeX platform with React, TypeScript, Firebase Auth/Firestore/Storage, BusyTeX WebAssembly compilation, PDF preview, templates, and project management.
- Implemented client-side E2EE using RSA-OAEP user keys, AES-GCM project encryption, passphrase-protected vaults, per-user wrapped master encryption keys, and idle-session lock behavior.
- Designed encrypted Yjs collaboration over a Node/Express WebSocket relay; the server validates Firebase identity and project access while relaying only opaque encrypted updates.

Zcash Developer & Investigative Tools | *ADAPT Centre*

- Authored and published an open-source JavaScript RPC wrapper for Zcash to simplify node interaction, shielded/transparent data querying, and developer workflows.
- Built a Flask and React-based block explorer for local network testing, ledger analysis, transaction inspection, and investigative debugging.
- Created reusable tooling that made complex blockchain data easier to inspect, validate, audit, and integrate into research and engineering workflows.

Blockchain-Native PKI & Secure Communication Wrapper | *ADAPT Centre*

- Designed a decentralized Public Key Infrastructure by mapping X.509 certificates to native NFTs on the XRP Ledger.
- Built an encrypted communication wrapper using blockchain-backed identity verification to support secure communication workflows.
- Implemented certificate encoding, identity verification, and key-management flows to reduce reliance on centralized trust anchors.

Proof-of-Work++ | *Trinity College Dublin*

- Proposed and simulated a consensus variant designed to expose cryptographically verifiable block validation progress before final block discovery.
- Evaluated protocol behavior across network fairness, transparency, energy efficiency, and security dimensions for academic publication.

PUBLICATIONS

PoW++: An Energy Efficient Proof-of-Work Variant | V. Agarwal, H. Vardhan, A. Kumar and H. Tewari | The 7th International Conference on Blockchain Computing and Applications, Oct 2025. Accepted.

A Blockchain-Native PKI: Secure X.509 Certificate Management on the XRP Ledger | H. Vardhan, P. Hiranayachatri, A. Lele and H. Tewari | IEEE International Conference on Distributed Ledger Technologies, Nov 2025. Accepted.

EDUCATION

Trinity College Dublin | MSc in Computer Science *Sept 2023 - Sept 2024*

Specialization: Intelligent Systems, Cryptography, and Blockchain Technology.

National Institute of Technology Karnataka, Surathkal | BTech in Information Technology *July 2016 - June 2020*

GPA: 8.48 / 10.0